

4 Lecture 4: Use of the Fundamental Solution for Poisson's Equation

Use of the fundamental solution to solve the Poisson equation in $\mathbb{R}^n$.

4.1 Recall: convolution and the fundamental solution

Recall from Lecture 3 that

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y)g(y) dy, \quad (4.1)$$

and, for $n \geq 2$, the fundamental solution of the Laplacian is

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}}, & n \geq 3, \end{cases} \quad \alpha(n) = |B_1|. \quad (4.2)$$

Formally,

$$-\Delta \Phi = \delta_0. \quad (4.3)$$

Exercise 4.1 (Convolution identities). For suitable f, g, h , verify

$$f * g = g * f, \quad (4.4)$$

$$f * (g * h) = (f * g) * h. \quad (4.5)$$

In this notation,

$$\delta_0 * \varphi = \varphi. \quad (4.6)$$

Indeed,

$$(\delta_0 * \varphi)(x) = \int_{\mathbb{R}^n} \delta_0(x - y)\varphi(y) dy = \varphi(x).$$

Thus, formally,

$$\begin{aligned} -\Delta(\Phi * f)(x) &= -\Delta_x \int_{\mathbb{R}^n} \Phi(x - y)f(y) dy \\ &= \int_{\mathbb{R}^n} [-\Delta_x \Phi(x - y)]f(y) dy \\ &= \int_{\mathbb{R}^n} \delta_0(x - y)f(y) dy = f(x), \end{aligned} \quad (4.7)$$

so

$$-\Delta(\Phi * f) = f \quad \text{at least formally.} \quad (4.8)$$

4.2 A rigorous Poisson solver

We verify the preceding calculation rigorously when f is “nice.”

Theorem 4.2 (Solving Poisson's equation in $\mathbb{R}^n$). Suppose $f \in C_c^2(\mathbb{R}^n)$; that is, f is twice continuously differentiable and compactly supported. Define

$$u(x) := (\Phi * f)(x) = \int_{\mathbb{R}^n} \Phi(x - y)f(y) dy. \quad (4.9)$$

Then

1. $u \in C^2(\mathbb{R}^n)$;
2. $-\Delta u = f$ in $\mathbb{R}^n$.

See Evans, §2.2.1(b).

Proof of (i). **Remark.** We cannot apply the second derivative directly to Φ in an expression such as

$$\int_{\mathbb{R}^n} D^2\Phi(x-y)f(y) dy,$$

because $D^2\Phi$ is not integrable near $y = x$.

Notice that

$$u = \Phi * f = f * \Phi = \int_{\mathbb{R}^n} f(x-y)\Phi(y) dy. \quad (4.10)$$

Therefore, the derivatives can instead fall on the smooth compactly supported factor f :

$$D_x^2 u(x) = \int_{\mathbb{R}^n} D_x^2 f(x-y)\Phi(y) dy. \quad (4.11)$$

Here Φ is integrable on bounded domains, and the integral is effectively over a bounded set because f is compactly supported. Hence differentiation inside the integral is justified (for example by the Lebesgue dominated convergence theorem), and $u \in C^2$.

Difference-quotient verification. Let $e_i = (0, \dots, 0, 1, 0, \dots, 0) \in \mathbb{R}^n$. From (4.10),

$$\frac{u(x + he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \Phi(y) \frac{f(x + he_i - y) - f(x - y)}{h} dy. \quad (4.12)$$

As $h \rightarrow 0$, the quotient converges uniformly on the relevant bounded set to $f_{x_i}(x - y)$. Thus

$$u_{x_i}(x) = \int_{\mathbb{R}^n} \Phi(y) f_{x_i}(x - y) dy. \quad (4.13)$$

Similarly,

$$u_{x_i x_j}(x) = \int_{\mathbb{R}^n} \Phi(y) f_{x_i x_j}(x - y) dy, \quad (4.14)$$

and these derivatives are continuous. Therefore $u \in C^2$.

4.3 Proof that $-\Delta u = f$

Fix $x \in \mathbb{R}^n$. By (4.14),

$$\Delta u(x) = \int_{\mathbb{R}^n} \Phi(y) \Delta_x f(x - y) dy. \quad (4.15)$$

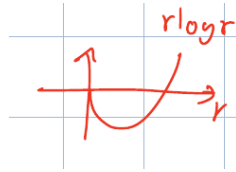
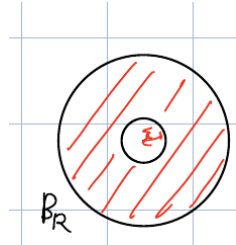
For $\varepsilon > 0$, split

$$\Delta u(x) = \underbrace{\int_{\mathbb{R}^n \setminus B_\varepsilon} \Phi(y) \Delta_x f(x - y) dy}_{I_\varepsilon} + \underbrace{\int_{B_\varepsilon} \Phi(y) \Delta_x f(x - y) dy}_{II_\varepsilon}. \quad (4.16)$$

For the second term,

$$|II_\varepsilon| \leq \sup_{\mathbb{R}^n} |\Delta f| \int_{B_\varepsilon} |\Phi(y)| dy \longrightarrow 0 \quad (\varepsilon \downarrow 0). \quad (4.17)$$

For $n = 2$ the radial integral is of order $\varepsilon^2 |\log \varepsilon|$, while for $n \geq 3$ it is of order ε^2 .

Figure 4.1: Original source sketch used in the small- ε estimate.Figure 4.2: Original source sketch: choose B_R large enough to contain the translated support.

Since $\Delta_x f(x - y) = \Delta_y f(x - y)$, choose $R \gg 1$ so that B_R contains the support of $y \mapsto f(x - y)$. Then

$$I_\varepsilon = \int_{B_R \setminus B_\varepsilon} \Phi(y) \Delta_y f(x - y) dy. \quad (4.18)$$

Recall Green's formula, for $g, h \in C^2$,

$$\int_{\Omega} (\Delta g) h = \int_{\Omega} g \Delta h + \int_{\partial \Omega} (h \partial_\nu g - g \partial_\nu h) dS. \quad (4.19)$$

Apply it to $\Omega = B_R \setminus \overline{B_\varepsilon}$, $g(y) = f(x - y)$, and $h(y) = \Phi(y)$. Since $\Delta \Phi = 0$ away from 0, and the outer-boundary terms vanish by the choice of R ,

$$\begin{aligned} I_\varepsilon &= \int_{\partial B_\varepsilon} \Phi(y) \partial_\nu f(x - y) dS(y) - \int_{\partial B_\varepsilon} f(x - y) \partial_\nu \Phi(y) dS(y) \\ &=: III_\varepsilon + IV_\varepsilon. \end{aligned} \quad (4.20)$$

Here ν is the outward unit normal for the punctured domain, so on the inner boundary it points toward the origin.

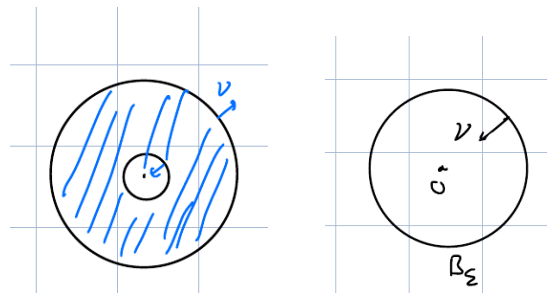


Figure 4.3: Original source sketches for the Green's-formula argument and the inner-boundary normal.

For the first boundary term,

$$|III_\varepsilon| \leq \sup |Df| \int_{\partial B_\varepsilon} |\Phi(y)| dS(y) \longrightarrow 0. \quad (4.21)$$

Indeed, the last integral is of order $\varepsilon |\log \varepsilon|$ if $n = 2$ and of order ε if $n \geq 3$.

For the second term, on the inner boundary,

$$\partial_\nu \Phi(y) = \begin{cases} \frac{1}{2\pi\varepsilon}, & n = 2, \\ \frac{1}{n\alpha(n)\varepsilon^{n-1}}, & n \geq 3, \end{cases} = \frac{1}{|\partial B_\varepsilon|}. \quad (4.22)$$

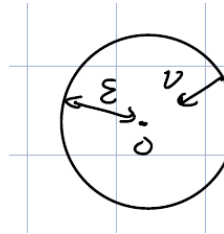


Figure 4.4: Original source sketch indicating the normal ν on ∂B_ε .

Therefore,

$$IV_\varepsilon = -\frac{1}{|\partial B_\varepsilon|} \int_{\partial B_\varepsilon} f(x-y) dS(y) \longrightarrow -f(x) \quad (\varepsilon \downarrow 0), \quad (4.23)$$

because f is continuous. Combining (4.17), (4.21), and (4.23),

$$\Delta(\Phi * f)(x) = -f(x),$$

i.e.

$$-\Delta(\Phi * f) = f. \quad (4.24)$$

This proves Theorem 4.2.

Next lectures.

- Fourier transform and PDE (Evans, §2.3.1).
- Solution to the heat equation.